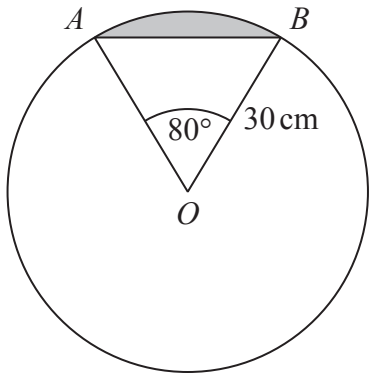
**15**




*AB* is a chord of a circle centre *O*.

The radius of the circle is 30 cm.
Angle *AOB* = 80°

Work out what percentage of the area of the circle is shaded.

| Angle | Area ($cm^2$) |
|---|---|
| 360 | $900\pi$ |
| 80 | $200\pi$ |

Circle = $900\pi$ $cm^2$.

Sector = $200\pi$ $cm^2$.

**1 mark**
80/360 of circle

Triangle AOB = ½ab sin C

= ½ × $30^2$ × sin 80° ≈ 443.1635 $cm^2$.

**1 mark**
Triangle area

Segment = $200\pi$ − 443.1635... ≈ 185.155 $cm^2$.

**1 mark**
Sector minus triangle

Percentage = 185.1550... ÷ ($900\pi$) × 100

**1 mark**
Part ÷ circle × 100

≈ 6.55%

**1 mark**
6.5%–6.6%

..... %

**(Total for Question 15 is 5 marks)**